Eric Maniraguha

Data Engineer · Database & Analytics Engineer · Data Engineering Consultant

Project Contributions & Technical Portfolio

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SQL PostgreSQL Oracle PL/SQL MySQL Python PySpark Apache Airflow dbt Kafka Power BI Grafana Docker AWS

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Professional Summary

Data Engineer and Database/Analytics professional with **10+ years of experience** designing data systems, ETL/ELT pipelines, relational databases, BI solutions, and digital transformation initiatives across **fintech, healthcare, humanitarian, research, and public-sector** environments. **MSc in Information Technology (Data Science & Machine Learning) from Carnegie Mellon University Africa.**

I help organizations turn fragmented operational data into **reliable, governed, decision-ready systems**, covering the full lifecycle:

> **Data modelling → Pipeline design → Data quality & reconciliation → Performance tuning → Dashboards & monitoring → Documentation & handover**

Impact at a Glance

Result	Context
98% data consistency across integrated datasets	Validation, reconciliation & quality controls, Dussoft Engineering
23% improvement in operational efficiency	Automated pipeline scheduling, error handling & logging, Dussoft Engineering
13% reduction in data processing time	Pipeline automation & performance tuning, Dussoft Engineering
90+ operational units supported	Financial & business database systems, BrightSeed Hub
18,200-record Golden Employee Dataset; **30** potential ghost employees flagged	HR Data Integration Pipeline (entity resolution & data quality)
8 business domains unified in one ERP data platform	Sales, inventory, HR, payroll, finance, assets, bookings, customers, DRA VANUA

Confidentiality: Why Most Code Lives Elsewhere

Most of my professional work was delivered **for client and partner organizations**. Their source code, data, and repositories are **private and owned by them**, so I don't link to or republish them. This repository presents that work responsibly:

✅ What you'll find here	❌ What you won't find here
Case studies: context, problem, my role, architecture, outcome	Proprietary source code or private repository links
Sanitized architecture and data-flow diagrams	Real customer, patient, or financial data
Original, runnable SQL I wrote to demonstrate the same techniques	Credentials, connection strings, or internal hostnames
Links to my own **public** project repositories	Client-confidential metrics or documents
Multi-year GitHub contribution history	

> 💬 **Happy to go deeper in an interview:** live walkthroughs of designs, schema decisions, and trade-offs, plus references from the organizations involved, on request.

Career-to-Project Map

How each role on my CV connects to the evidence in this repository.

Period	Organization	Role	Evidence in this repo
Nov 2025 – Present	**BrightSeed Hub Ltd.**	Database Engineering & Systems Analyst Consultant	Case study

Period	Organization	Role	Evidence in this repo
Jul 2026 – Oct 2026	**eFiche**	Digital Health Implementation Consultant	Case study
2026	**DRA VANUA Group Ltd**	ERP Data Engineering & Systems Development (project lead)	Case study · Repository
May 2023 – Nov 2025	**Dussoft Engineering**	Data Engineering & Database Systems Consultant / Technical Focal Person	Case study
Jun 2023 – Dec 2025	**CyLab Africa / Upanzi Network**	Research Associate: Data Scientist & ICT Systems Analyst	Upanzi Mojaloop · CyLab Africa
Aug 2023 – Dec 2023	**C4IR Rwanda**	AI Developer	AI Document Navigator
Jul 2019 – Sep 2022	**Dussoft Engineering**	Data Engineer & Team Lead	Humanitarian Market Intelligence
Apr 2015 – May 2019	**Tech-Connect Business Park**	ICT Systems & E-Business Solutions Consultant (Lusaka)	Case study
Sep 2024 – Present	**AUCA** (and UNILAK, University of Kigali)	Assistant Lecturer: Oracle PL/SQL, Big Data Analytics, MIS	Code samples used as teaching-grade examples

Project Case Studies

Each case study follows the same consulting format: **Context → Problem → What I Delivered → Architecture → Impact → Technologies.**

💳 Fintech & Payments

Project	My Data Engineering Contribution	Core Technologies
Upanzi Network: Mojaloop Digital Public Infrastructure · payment switch	Multi-currency ledger data model; reporting dashboards; Grafana observability & log tracking; SQL optimization and indexing	PostgreSQL, MySQL, Grafana, Loki, Prometheus, Node.js
FinTech Core & Payment Engine Financial services · payments	Double-entry ledger schema; automated settlement **reconciliation pipelines**; data integrity controls	PostgreSQL, Redis, RabbitMQ/BullMQ, Docker

Project	My Data Engineering Contribution	Core Technologies
BrightSeed Hub Financial & business applications · 90+ units	Schema, stored procedure & index optimization; migration, cleansing & reconciliation; automated data quality checks; Grafana monitoring; RBAC, backup & recovery	PostgreSQL, SQL, Python, Grafana

🏥 Healthcare

Project	My Data Engineering Contribution	Core Technologies
eFiche: EHR Data Migration Digital health · electronic health records	Led data migration & database implementation; validation, transformation & integrity checks; SQL troubleshooting; ETL and interoperability support; English–French localization	SQL, ETL, data validation, REST APIs
GeoScalable Malaria Surveillance Kamabuye Health Center · 2024–2025	Geospatial data pipelines and GIS dashboards for malaria hotspot analysis	MongoDB, PostgreSQL, Airflow, Django, Elasticsearch, Kibana

🏛️ Public Sector, Research & AI

Project	My Data Engineering Contribution	Core Technologies
CyLab Africa Cybersecurity & DPI research	Security **telemetry ingestion pipeline**; event categorization for threat analytics; data platform administration and governance	Python, PySpark, PostgreSQL, Elasticsearch, CI pipelines
AI-Powered Document Navigator C4IR Rwanda · government information retrieval	NLP/LLM ingestion pipeline: cleansing, chunking, embeddings; vector storage, metadata & similarity search; RAG over REST APIs	Python, LLMs, vector databases, REST APIs

🏢 Enterprise, ERP & Supply Chain

Project	My Data Engineering Contribution	Core Technologies
DRA VANUA ERP Platform Led end-to-end · 2026	MySQL data platform integrating 8 business domains; Sequelize migrations; RBAC & audit logging; ETL, BI & KPI dashboards; Grafana monitoring; Dockerized deployment	MySQL, Node.js, Sequelize, Grafana, Docker

Project	My Data Engineering Contribution	Core Technologies
Dussoft Engineering Technical focal person · 2023–2025	Oracle/PostgreSQL/MySQL administration; Airflow ETL with validation & reconciliation; migration, backup & recovery; runbooks; Power BI & Kibana dashboards	Oracle, PL/SQL, PostgreSQL, MySQL, Python, Airflow, Power BI, Kibana
Humanitarian Market Intelligence Dussoft · 2019–2022	Multi-source ETL/ELT into PostgreSQL; dedup & schema validation; automated quality checks; food-price and supply-chain Power BI dashboards	PostgreSQL, Python, Pandas, Airflow, AWS S3, Redshift, Power BI
Tech-Connect Business Park Lusaka · 2015–2019	Database design & administration; e-payment and transaction workflows; RBAC data access design; system integration and data quality controls	SQL, relational databases, e-payment integrations

Public Repositories

Projects I own end-to-end, with source code available.

Repository	What it demonstrates	Highlights
DRA VANUA ERP	Full ERP data platform: schema design, migrations, RBAC, audit logging, Grafana	8 integrated domains, Dockerized, automated DB migrations
HR Data Integration Pipeline	Python ETL over CSV, JSON, XLSX & XML; schema harmonization; entity resolution	18,200-record golden dataset, 30 ghost employees flagged, automated tests
Humanitarian Market Dashboard	Published Power BI results	Food-price monitoring & market trends

Code Samples

Because client code can't be shared, these samples were **written from scratch** to demonstrate the techniques I apply on client projects and teach in Oracle PL/SQL and Advanced Database courses. All target **PostgreSQL 15+**.

#	Section	What it demonstrates	Technologies
01	**Database Administration**	Schema design with constraints, views & materialized views, stored procedures, functions, audit triggers, role hierarchy, **row-level security**, monitoring queries	PostgreSQL, PL/pgSQL
02	**Performance Tuning**	End-to-end slow-query case study: identify → `EXPLAIN ANALYZE` → index → rewrite → verify; CTEs, window functions, database statistics	PostgreSQL, pg_stat_statements

Quick start

```
createdb portfolio_demo
psql -d portfolio_demo -f code-samples/01-database-administration/schemas/<schema-file>.sql
```

Skills Alignment

Competency	Where it's shown
Data modelling (relational, ledger, multi-tenant, ERP)	Upanzi Mojaloop, FinTech ARCHITECTURE.md, DRA VANUA ERP, schemas/
ETL/ELT pipelines & orchestration	Dussoft Engineering, Humanitarian Market Intelligence, CyLab telemetry pipeline
Data migration & reconciliation	eFiche EHR migration, FinTech reconciliation workers, BrightSeed Hub
Data quality & entity resolution	HR Data Integration Pipeline, triggers & constraints
Query optimization & performance	02-performance-tuning, Upanzi reporting optimization
Analytics, BI & dashboards	Humanitarian dashboards, Upanzi Grafana, DRA VANUA KPI reporting
Data security & access control	roles-and-permissions/, FinTech compliance doc
Monitoring & observability	monitoring/, BrightSeed Grafana, Upanzi Grafana & Loki
AI/ML & LLM data pipelines	C4IR Document Navigator, Malaria surveillance
Documentation & handover	Every case study and section README; runbooks described in Dussoft Engineering
Sustained hands-on engineering	GitHub contribution history

Technical Skills

Area	Tools & Practices
Databases	PostgreSQL, Oracle & PL/SQL, MySQL, SQL Server, MongoDB, Elasticsearch, Amazon Redshift: modelling, procedures, triggers, indexing, query optimization, capacity planning
Data Engineering	Python, PySpark, Pandas, ETL/ELT, Apache Airflow, Kafka, dbt, data lakes, data warehousing
Reliability & Operations	Monitoring, health checks, backup & recovery, DR, replication, migration, patching, incident troubleshooting, runbooks

Area	Tools & Practices
Governance & Security	RBAC, row-level security, auditing, data quality, metadata management, compliance controls
Monitoring & BI	Grafana, Kibana, Prometheus, Loki, Power BI, SQL analytics
AI & ML	Scikit-learn, TensorFlow, deep learning, LLMs, RAG, embeddings, vector databases
Cloud & DevOps	AWS (S3, Redshift), Docker, Kubernetes, CI/CD, MLOps
GIS	ArcGIS Pro, Google Earth Engine, spatial analytics

Education & Certifications

Education

- **MSc Information Technology (Data Science & Machine Learning)**, Carnegie Mellon University Africa, 2021–2024 · GPA 3.60
- **BSc Information Management**, Chreso University (with North End State University College), 2013–2017 · Merit

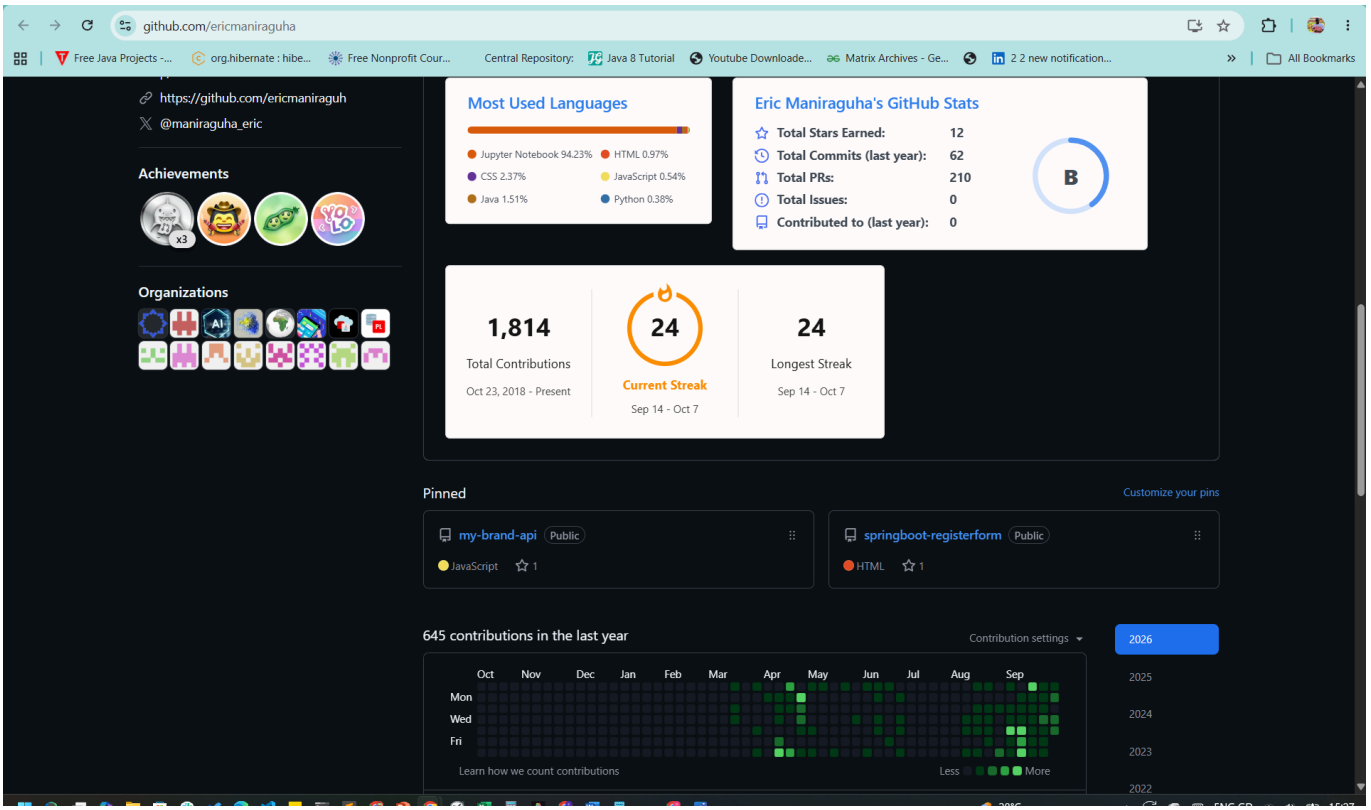
Certifications

- Data Engineer Certification, DataCamp
- SQL Associate Certification, DataCamp (valid to Sep 2028)
- Big Data Fundamentals with PySpark, DataCamp (Oct 2026)
- Kubernetes and Cloud Native Associate (KCNA), Linux Foundation
- Full-Stack Developer Certification, Andela Rwanda
- 15+ additional certifications in AI/ML, data governance, databases, statistics, and FinTech

GitHub Contribution History

Much of my day-to-day work happens in **private organizational repositories**, so these graphs show activity that the public repositories alone don't reflect.

▼ 2026



▶ **2025**
▶ **2024**
▶ **2023**

Evidence & Sanitization Policy

- **Code samples** in `code-samples/` are my **original work**, written for this portfolio.
- **Case studies** in `case-studies/` describe work delivered for organizations. Diagrams and screenshots are **sanitized**: no proprietary code, credentials, personal data, or confidential identifiers.
- **Metrics** quoted here are the same figures stated on my CV and can be discussed in detail at interview.
- Organization names are used only to give context for my role. All rights to the underlying systems belong to their owners.

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